To the attention of Mr. Gaël Veyssière, Ambassador of France to Croatia, on behalf of Professor Miroslav Radman

The project of a bilateral scientific collaboration, of strategic importance,

between France and Croatia

Active prevention against the pathogenic effects of radiation and aging

## Motivation

The motivation for this project is the use of the results resulting from the collaboration between French and Croatian researchers, woven for more than 15 years by Pr Miroslav Radman to try to solve some of the major societal problems related to the harmful effects of radiation and of ageing. This collaboration is summarized at the end of the text with the corresponding references.

We don't have time for trivialities, no matter how high-tech. The constellation is such that we need “miracles”. Herman Melville said, "I'd rather not catch a big fish than catch a small one. »

## The societal and political context

The current geopolitical situation presents the increased risk of the use of nuclear weapons with well-known biological effects of radiation since the atomic bombs on Hiroshima and Nagasaki. Even without war, the need for nuclear power plants and the risk of accidents is a permanent danger for workers and the population (example of Chernobyl). Also, space flights are moving into the commercial phase, yet the effects of radiation on the health of astronauts are much more serious than the general perception.

## The scientific context

Our research, carried out in France and Croatia, on organisms extremely resistant to radiation, such as the bacterium Deinococcus radiodurans and small animals such as bdelloid rotifers (collaboration with Pr. M. Meselson of Harvard), brings real hope of be able to neutralize the biological effects of ionizing and ultraviolet (UV) radiation. These organisms have a 100% survival rate - and this without detectable effects (including genetic mutations) - at radiation doses a thousand times higher than the lethal dose for humans (see annexes).

It turns out that the mechanism of extreme biological radioresistance is simple and identical to that which ensures longevity and prevents the emergence of malignant and degenerative diseases. Thus, our basic research opens up an avenue of application that promises the maintenance of human health in the face of radiation (radioresistance) and time (chronoresistance).

## The concept

There is a small fraction of the human population (4 in 10,000) that is resistant to all diseases simultaneously and therefore can live beyond 100 years – centenarians and supercentenarians. Their resistance to radiation has not yet been studied, but their existence testifies to the real possibility of being able to simultaneously eliminate almost all fatal diseases (cause of approximately 90% of mortality in Europe and the USA) and would thus be a real revolution. of medicine. If successful, this revolution would reduce the cost of public health between ten and a hundred times. Currently, our conclusion is that there is a common cause for all fatal diseases and that we can act on this common cause to prolong healthy life and protect ourselves from the harmful effects of radiation.

## A common mechanism for health, longevity and resistance to radiation

In 2010, two articles, one by the group of M. Daly (USA) and the other by A. Krisko and M. Radman, demonstrated that the presence of small molecular complexes that protect proteins against corrosion (damage oxidative) is responsible and sufficient to ensure the extreme resistance of the bacterium Deinococcus radiodurans. In 2021, Daly then demonstrated that this same antioxidant complex prolongs the life of a small worm, the nematode C. elegans. In our institute, the NAOS Group team found, in another bacterium as radioresistant as D. radiodurans, another small molecule with antioxidant chaperone properties (meaning that it binds to proteins). We can therefore hope to find, in highly radioresistant organisms, a multitude of natural molecules that protect proteins against oxidative damage.

These antioxidant chaperone molecules are non-toxic, making them prime candidates for the prevention of the effects of irradiation, but also of all age-related diseases. We thus propose the concept that there is "a single disease" linked to aging which is divided in the human population into a hundred diseases according to the innate individual predispositions ("weak links"). This mechanism is known to us (see Krisko and Radman, 2019). Antioxidant chaperones bind and further protect proteins weakened by mutation (polymorphism, which predisposes to a particular disease).

In summary :

(1) Protein oxidation disrupts and reduces cellular activity and exhibits “chemical noise” that increases with age (aging).

(2) Excessive oxidation of a particular protein (made fragile by a benign mutation called a polymorphism) leads to a particular disease predisposed by the mutation.

(3) If the mutation is severe (id est, directly destroys the function of the protein), the disease is already manifested at birth – then it is a syndrome.

(4) Often, the oxidized protein becomes "sticky", it aggregates, becomes toxic for the cell and leads to a degenerative disease, such as Alzheimer's, Parkinson's or Charcot's disease (ALS). If it does not aggregate, it becomes non-functional or mal-functional and leads to cancer.

(5) Since the chemistry of aging and predestined disease is protein corrosion, the prevention of protein corrosion should prevent or neutralize the biological effects of radiation and simultaneously prevent all age-related diseases.

Franco-Croatian collaboration

Thanks to the dual Franco-Croatian nationality of the founder and scientific director of the Mediterranean Institute of Life Sciences (MedILS) in Split, Professor Miroslav Radman, the collaboration becomes quite natural. Former students of Miroslav Radman, Drs. François Taddei, Ivan Matic and Ariel Lindner were heavily involved in starting MedILS.

The MedILS is an international scientific institution, independent and private, but without personal profit, which enjoys a status similar to that of the Curie and Pasteur institutes. In 2004, the MedILS building was rehabilitated and made available to the founders (40-year concession by the government), professors Miroslav Radman and Marija Alacevic. MedILS has been operational since 2007. The Croatian government does not yet participate in the financing of MedILS.

Currently, funding comes exclusively from private French and English sources; the most important being that of a French industry and a generous man, Mr. Jean-Noël Thorel, founder of the cosmetic companies Bioderma, Esthederm and Etat Pur which form the NAOS group. Mr. Thorel has introduced biology into cosmetics and is particularly interested in skin aging. On the other hand, MedILS is the first institute in continental Europe declared as a research institute on the biology of aging and age-related diseases. It is thanks to Mr. Thorel and his support that the MedILS exists today.

In 2021, MedILS signed a collaboration agreement with Institut Curie (where Prof. Radman sat for a decade as a member of the scientific council), as well as with the University of Split in 2022. We are also successfully pursuing close collaborations with French colleagues and friends, in particular with Dr. Laurent Schwartz, oncologist from the Assistance Publique des Hôpitaux de Paris, as well as with Pr. Jean-Marc Lemaitre, director of the Functional Genomics Institute of the INSERM in Montpellier and expert in cellular reprogramming and aging.

Miroslav Radman is professor emeritus, of exceptional class, of cell biology at the Faculty of Medicine of the University R. Descartes (Paris-5). Since 2002, he has been a member of the Institut de France – Académie des Sciences. He is also a member of 8 international and national academies including the National Academy of Sciences (USA) and the American Academy of Arts and Sciences; he is a member of the scientific council of the IHU infection of Marseille as well as of the Pasteur Institute of Lille. Miroslav Radman is Chevalier of the Legion of Honor.

The intention: to create a “miracle”

The international geopolitical situation does not arouse optimism. The use of nuclear weapons is the sword of Damocles hanging over our heads. Nuclear energy, with its risks of accident, is currently a dangerous necessity. Commercial space flights are gearing up. It is difficult to imagine how to protect the population – and even individuals – from the harmful biological effects of ionizing radiation. The exception is the research and development (R&D) offered here.

On the other hand, the cost of public health is increasing exponentially, especially because of the prices of increasingly sophisticated treatments which have a common characteristic: they do not cure diseases (even when they are called "smart drugs" smart).

On the basis of research already carried out by a free and spontaneous Franco-Croatian collaboration, led by Professor Miroslav Radman for 15 years, we propose here to recognize this collaboration by the respective governments and provide the means necessary to solve the two problems: biological radiation protection and the eradication of age-related diseases (that is, to make us all centenarians!). These are two major projects of strategic importance to be carried out quickly under the high patronage of the two governments. The economic, psychological and even military impact of the success of these two projects on the whole world would be unimaginable. We do not want the powerful lever of power of a "knowledge monopoly" to end up as private property. Rather, we see France and Croatia coming closer and emerging together as leaders of a healthier and safer world than now.

To keep control of the application, we ask the French and Croatian governments to ensure the support and extension of the existing Franco-Croatian collaboration on these two major strategic projects:

(1) “Radioduran” project - Preclinical tests of natural substances that protect against the biological and genetic effects of radiation.

(2) “Chronoduran” project - Preclinical testing of molecules that slow down, stop or reverse the aging process and prevent or reverse malignant and degenerative diseases associated with aging.

Financial support for research is initially planned for the MedILS institute in Croatia and for the Institut Curie in France, with a lasting alliance of our closest and most important partners, Mr Jean-Noël Thorel (founder of NAOS), Dr Laurent Schwartz (AP-HP oncologist) and Pr. Jean-Marc Lemaitre (INSERM, Montpellier). Other associates and partners, such as dermatologist Pr. Brigitte Dreno (INSERM, Nantes), mathematician, biologist and KOL Dr. Idriss Aberkane, ets., are expected over time. Cellular biologist, specialist in neurodegenerative diseases, Dr. Katarina Trajković will become director of research at MedILS. Prof. Matthew Meselson (Harvard), Dr. Robert Wagner (GeneCheck, Fort Collins, Colorado, USA) and Dr. Michael Daly (NIH, USA) will be called upon as scientific advisors on the projects.

- This text was developed with the collaboration of Dr Zoran Dermanović, member of the MedILS Board of Directors.